

Operational Properties 2

Theorem 7.4.1 Derivatives of transforms

if $F(s) = \mathcal{L}\{f(t)\}$ and $n = 1, 2, 3, \dots$, then $\mathcal{L}\{t^n f(t)\} = (-1)^n \frac{d^n}{ds^n} F(s)$

ex

evaluate $\mathcal{L}\{te^{-3} \cos 3t\}$ notice the t has a exponent equal to one so we are going to take the first derivative so the n becomes $(-1)^1$

$$\mathcal{L}\{te^{-3} \cos 3t\} = (-1)^1 \frac{d}{ds} \left[\mathcal{L}\{e^{-3t} \cos(3t)\} \right]$$

$$\left\{ \frac{s}{s^2 + 9} \right\} s \rightarrow s+3$$

$$\mathcal{L}\{te^{-3} \cos 3t\} = - \frac{d}{ds} \left[\frac{s+3}{(s+3)^2 + 9} \right]$$

$$\frac{F'g - Fg'}{g^2}$$

$$F = s+3$$

$$g = s^2 + 6s + 18$$

$$F' = 1$$

$$g' = 2s + 6$$

$$= \frac{(1)(s^2 + 6s + 18) - (s+3)(2s+6)}{(s^2 + 6s + 18)^2}$$

$$= - \left[\frac{-s^2 - 6s}{(s^2 + 6s + 18)^2} \right]$$

$$= \boxed{\frac{s^2 + 6s}{(s^2 + 6s + 18)^2}}$$

Convolution theorem

if f and g are piecewise continuous on $[0, \infty)$, then

$$f \cdot g = \int_0^t f(\tau) g(t - \tau) d\tau$$

$$\mathcal{L}\{f \cdot g\} = \mathcal{L}\{f(t)\} \mathcal{L}\{g(t)\} = F(s) G(s)$$

note : $f \cdot g = g \cdot f$

$$a) \mathcal{L}^{-1}\{F(s) G(s)\} = f \cdot g$$

$$b) \mathcal{L}\left\{\int_0^t f(\tau) d\tau\right\} = \frac{1}{s} F(s)$$

$$c) \int_0^t f(\tau) d\tau = \mathcal{L}^{-1}\left\{\frac{1}{s} F(s)\right\}$$

evaluate :

$$\mathcal{L}\left\{\int_0^t \tau \sin(\tau) d\tau\right\}$$

$$\Rightarrow \frac{1}{s} \mathcal{L}\{t \sin(t)\}$$

$$= \frac{1}{s} \left[-\frac{d}{ds} \mathcal{L}\{\sin(t)\} \right]$$

$$= -\frac{1}{s} \left[\frac{d}{ds} \left(\frac{1}{s^2 + 1} \right) \right]$$

$$= \boxed{\frac{2}{(s^2 + 1)^2}}$$

ex: evaluate $\mathcal{L}\left\{\int_0^t \sin(\tau) \cos(t-\tau) d\tau\right\}$

Soln: $\Rightarrow \mathcal{L}\{\sin(t)\} \mathcal{L}\{\cos(t)\} = \frac{7.1}{s}$
 $\Rightarrow \left[\frac{1}{s^2+1}\right] \left[\frac{s}{s^2+1}\right] = \frac{s}{(s^2+1)^2}$

ex: Evaluate $\mathcal{L}^{-1}\left\{\frac{1}{s^2(s-1)}\right\}$

Soln: $\mathcal{L}^{-1}\left\{\frac{1}{s} F(s)\right\} = \int_0^t f(\tau) d\tau$

Note:

So, $\mathcal{L}^{-1}\left\{\frac{1}{s-1}\right\} = e^t$

and $\mathcal{L}^{-1}\left\{\frac{1}{s(s-1)}\right\} = \int_0^t e^{\tau} d\tau = e^{\tau} \Big|_0^t = e^t - 1$

Thus, $\mathcal{L}^{-1}\left\{\frac{1}{s^2(s-1)}\right\} = \int_0^t (e^{\tau} - 1) d\tau$
 $= e^{\tau} - \tau \Big|_0^t = \boxed{e^t - t - 1}$

this is another couple of theorem we might have already included them, talked about them

Facts:

① $\mathcal{L}\left\{\int_0^t f(\tau) d\tau\right\} = \frac{1}{s} F(s)$

② $\int_0^t f(\tau) d\tau = \mathcal{L}^{-1}\left\{\frac{1}{s} F(s)\right\}$

③ $f(t) = g(t) + \int_0^t f(\tau) h(t-\tau) d\tau$

Volterra Integral Equation
Volterra

ex: Solve $t - 2f(t) = \int_0^t (e^{\tau} - e^{-\tau}) f(t-\tau) d\tau$

Soln.

$$\mathcal{L}\{t\} - 2\mathcal{L}\{f(t)\} = \mathcal{L}\{e^{\tau} - e^{-\tau}\} \mathcal{L}\{f(t)\}$$

$$\Rightarrow \frac{1}{s^2} - 2F(s) = \left[\frac{1}{s-1} - \frac{1}{s+1} \right] F(s)$$

Solve for $F(s)$

$$\Rightarrow \frac{1}{s^2} = F(s) \left[\frac{1}{s-1} - \frac{1}{s+1} + 2 \right]$$

$$\Rightarrow \frac{1}{s^2} = F(s) \left[\frac{s+1 - (s-1) + 2(s+1)(s-1)}{s^2-1} \right]$$

$$\Rightarrow \frac{1}{s^2} = F(s) \left[\frac{s^2}{s^2-1} \right] \Rightarrow F(s) = \frac{s^2-1}{2s^4}$$

$$\Rightarrow F(s) = \frac{1}{2} \frac{1}{s^2} - \frac{1}{2} \frac{1}{s^4}$$

$$\Rightarrow F(s) = \frac{1}{2} \frac{1}{s^2} - \frac{1}{2} \cdot \frac{1}{6} \cdot \frac{6}{s^4}$$

$$\Rightarrow \boxed{f(t) = \frac{1}{2}t - \frac{1}{12}t^3}$$

ex: Solve the Integro-differential eqn by L.T.
 $y'(t) - \frac{1}{2} \int_0^t (t-v)^2 y(v) dv = -t, y(0)=1$

Soln.

Take $\mathcal{L}$ to the eqn.

$$\Rightarrow \mathcal{L}\{y'\} - \frac{1}{2} \mathcal{L}\{t^2\} \mathcal{L}\{y(t)\} = -\mathcal{L}\{t\}$$

$$\Rightarrow sY(s) - y(0) - \frac{1}{2} \frac{2}{s^3} \cdot Y(s) = -\frac{1}{s^2}$$

$$\Rightarrow sY(s) - 1 - \frac{1}{s^3} Y(s) = -\frac{1}{s^2}$$

$$\Rightarrow Y(s) \left[s - \frac{1}{s^3} \right] = 1 - \frac{1}{s^2}$$

$$\Rightarrow Y(s) = \frac{s^2-1}{s^2} \cdot \frac{s^3}{(s^2+1)(s^2-1)}$$

$$\Rightarrow Y(s) = \frac{s}{s^2+1}$$

Hence, $\mathcal{L}^{-1}\{Y(s)\} = \mathcal{L}^{-1}\left\{ \frac{s}{s^2+1} \right\}$

$$\Rightarrow \boxed{y(t) = \cos t}$$